

Failure of a pairwise four-step bound for restarted Anderson acceleration

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Independently reviewed resolution. The complete proof was checked against the current canonical target IE-18. The recovered provisional identifiers are confirmed. Authorship is recorded at the submitter's request; substantial AI assistance and reconstruction are disclosed in the submission record. The dated independent agent review records the subsequent checks and supersedes original draft remarks about pending review. This is not external human peer review or formal certification.

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

Exact diagonal examples refute a proposed pairwise formula for four-step residual amplification. One example has both the iteration matrix and its complement positive definite. A positive-definite one-parameter family gives an unbounded ratio of actual amplification to the proposed expression. These finite-step statements do not resolve the separate asymptotic convergence-factor conjecture.

1 The finite-step assertion

For symmetric M with $A = I - M$ nonsingular, define

$$\alpha(v) = \frac{v^T A v}{v^T A^2 v}, \quad R(v) = M[I - \alpha(v)A]v \quad (v \neq 0), \quad R(0) = 0.$$

The residual after four original steps is $R(R(v))$. For eigenvalues m_i , let

$$\Lambda(M) = \max_{i < j} \left(\frac{m_i m_j (m_j - m_i)}{|m_i(1 - m_i)| + |m_j(1 - m_j)|} \right)^2.$$

For the formula to be defined, each pair considered must have nonzero denominator; all pairs in the examples below do. The target assertion as recovered is $\max_{v \neq 0} \|R(R(v))\|_2 / \|v\|_2 = \Lambda(M)$. The square is part of the proposed norm amplification; squared-norm comparisons require $\Lambda(M)^2$.

2 Exact examples

For $M = \text{diag}(1/10, 1/2, 3/5)$ and $v = (1, 1, 1)^T$, direct rational substitution gives

$$\alpha(v) = 90/61, \quad R(v) = (-2, 8, 15)^T/61, \quad \alpha(R(v)) = 3140/1381,$$

$$R(R(v)) = (289, -756, 1125)^T/84241, \quad \Lambda(M) = 1/121.$$

Consequently

$$\frac{\|R(R(v))\|_2^2}{\|v\|_2^2} = \frac{1920682}{21289638243} > \frac{1}{14641} = \Lambda(M)^2,$$

because $1920682 \cdot 14641 = 28120705162 > 21289638243$. Both M and $I - M$ are positive definite, and $\rho(M) = 3/5 < 1$.

For the simpler semidefinite $M = \text{diag}(0, 1/2, 2/3)$,

$$\alpha(v) = 66/49, \quad R(v) = (0, 8, 18)^T/49, \quad \alpha(R(v)) = 35/13,$$

$$R(R(v)) = (0, -18, 16)^T/637, \quad \Lambda(M) = 4/289.$$

Here the squared comparison is $580/1217307 > 16/83521$, certified by $48442180 > 19476912$ after cross multiplication. No optimization or tolerance is involved.

3 Unbounded underestimation

Let $M_\varepsilon = \text{diag}(\varepsilon^2, 1/2, 1/2 + \varepsilon)$, $v = (1, 1, 1)^T$, with $0 < \varepsilon < 1/2$. Both M_ε and its complement are positive definite. We claim

$$\frac{\|R_\varepsilon(R_\varepsilon(v))\|_2}{\|v\|_2} = \frac{\varepsilon}{6\sqrt{6}} + O(\varepsilon^2), \quad \Lambda(M_\varepsilon) = \frac{\varepsilon^2}{4} + O(\varepsilon^3).$$

All intermediate denominators are nonzero at $\varepsilon = 0$. Expansion gives

$$\alpha_\varepsilon(v) = \frac{4}{3} + \frac{2}{9}\varepsilon + O(\varepsilon^2).$$

For $w_\varepsilon = R_\varepsilon(v)$,

$$w_0 = (0, 1/6, 1/6)^T, \quad w'_0 = (0, -1/18, 17/18)^T.$$

With $a(\varepsilon) = (1 - \varepsilon^2, 1/2, 1/2 - \varepsilon)$, the second coefficient is

$$\beta_\varepsilon = \frac{\sum_i a_i w_{\varepsilon,i}^2}{\sum_i a_i^2 w_{\varepsilon,i}^2} = 2 + 2\varepsilon + O(\varepsilon^2).$$

Its numerator has value $1/36$ and derivative $13/108$ at zero, and denominator has value $1/72$ and derivative $5/108$. Differentiating the second residual gives

$$R_\varepsilon(w_\varepsilon) = \varepsilon(0, -1/12, 1/12)^T + O(\varepsilon^2),$$

proving the norm expansion. The pair $1/2, 1/2 + \varepsilon$ contributes

$$\left(\frac{\varepsilon(1 + 2\varepsilon)}{2 - 4\varepsilon^2} \right)^2 = \varepsilon^2/4 + O(\varepsilon^3).$$

Pairs involving ε^2 contribute $O(\varepsilon^4)$, since their denominators tend to $1/4$. Thus the former pair is maximal for sufficiently small ε , proving the second expansion. Their ratio is

$$\boxed{\frac{2}{3\sqrt{6}}\varepsilon^{-1} + O(1) \longrightarrow \infty.}$$

No fixed multiplicative constant repairs the proposed expression as a uniform four-step upper bound on positive-definite contractions.

4 Independent recurrence check and limits

The code independently executes two cycles of

$$r_1 = Mr_0, \quad d = r_1 - r_0, \quad \gamma = -r_1^T d / (d^T d), \quad r_2 = M(r_1 + \gamma d).$$

Since $d = -Ar_0$, this gives $r_2 = R(r_0)$. The exact vectors and inequalities are checked in both formulations. The continuous-family result is analytic, not established by finite sampling. A large amplification at one residual need not persist on an orbit; the asymptotic-convergence conjecture remains outside these results.

5 Sources and scope

Recovered references: OpenProblemsInNLA, provisionally IE-18, *Exact four-step amplification for restarted Anderson acceleration*; O. A. Krzysik, H. De Sterck and A. Smith, *Asymptotic convergence of restarted Anderson acceleration for certain normal linear systems*, SIAM J. Sci. Comput. 47 (2025), DOI 10.1137/24M1672262, arXiv:2312.04776v4, equation (27).¹ The finite-step assertion in Section 1 is the precise mathematical target here. No assertion is made that a different formula, exponent convention, or asymptotic statement in the original paper is refuted without verifying that source.

¹<https://arxiv.org/html/2312.04776v4>. Bibliographic details and exact target scope are retained as source leads, not independently verified during reconstruction.